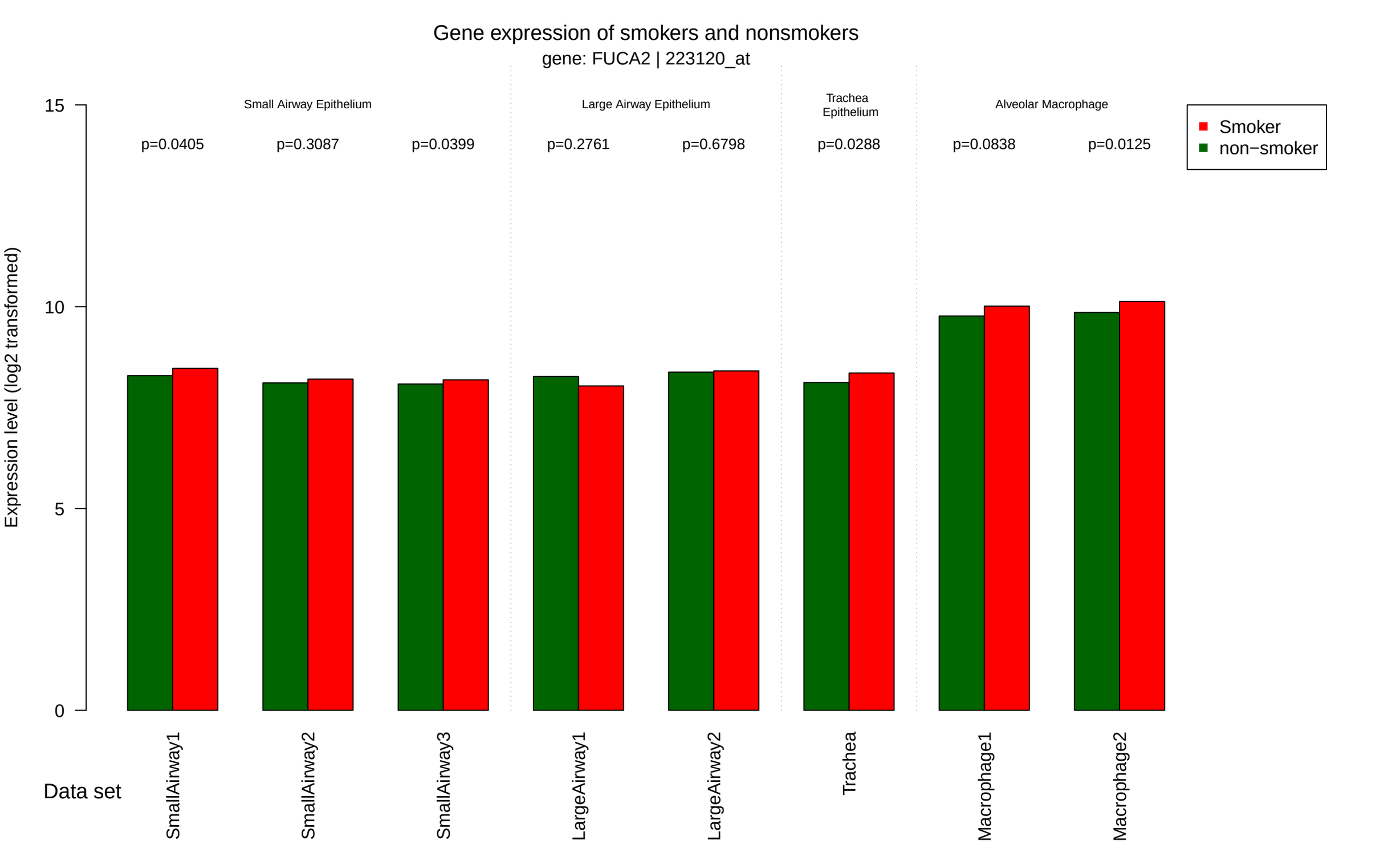
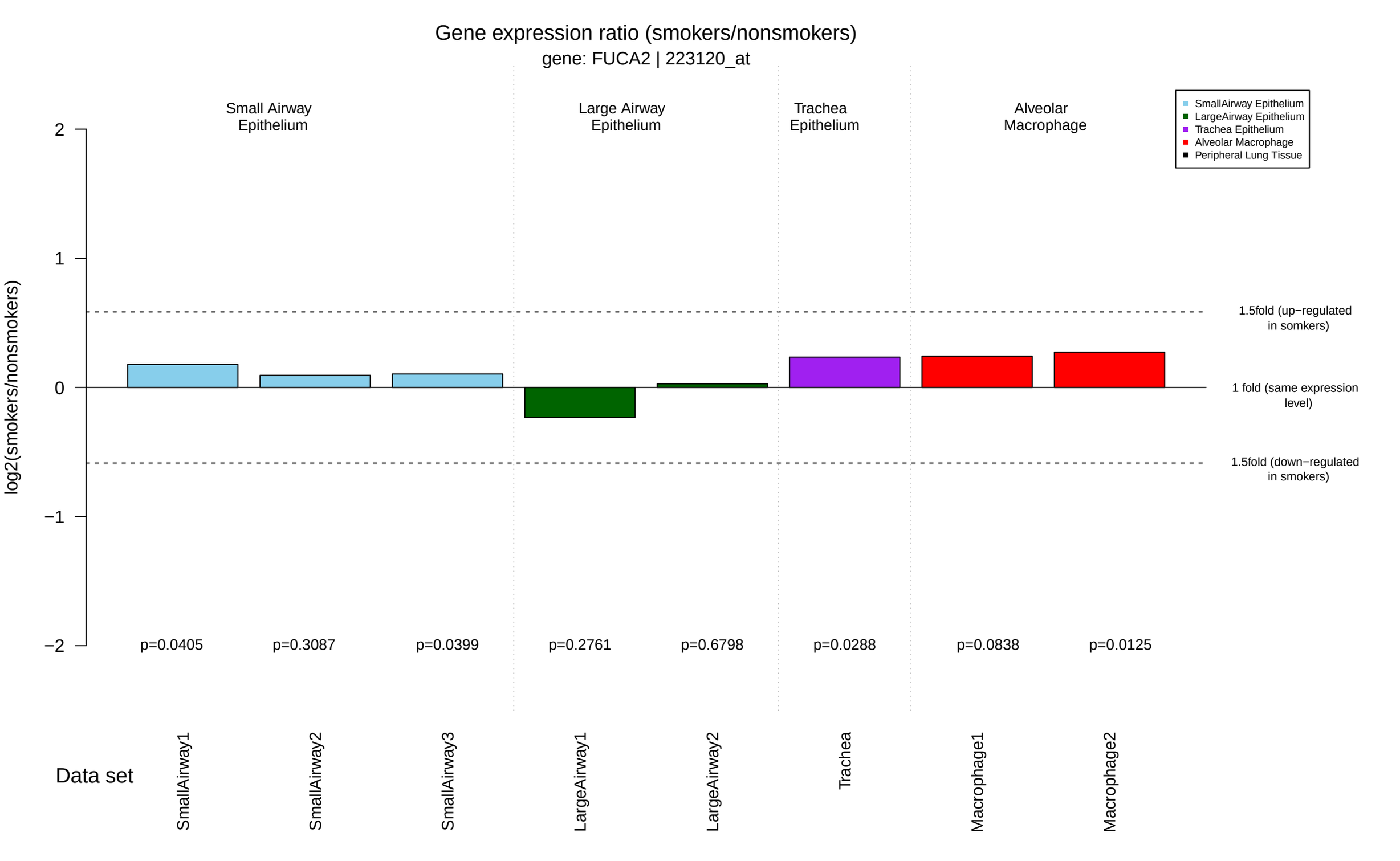
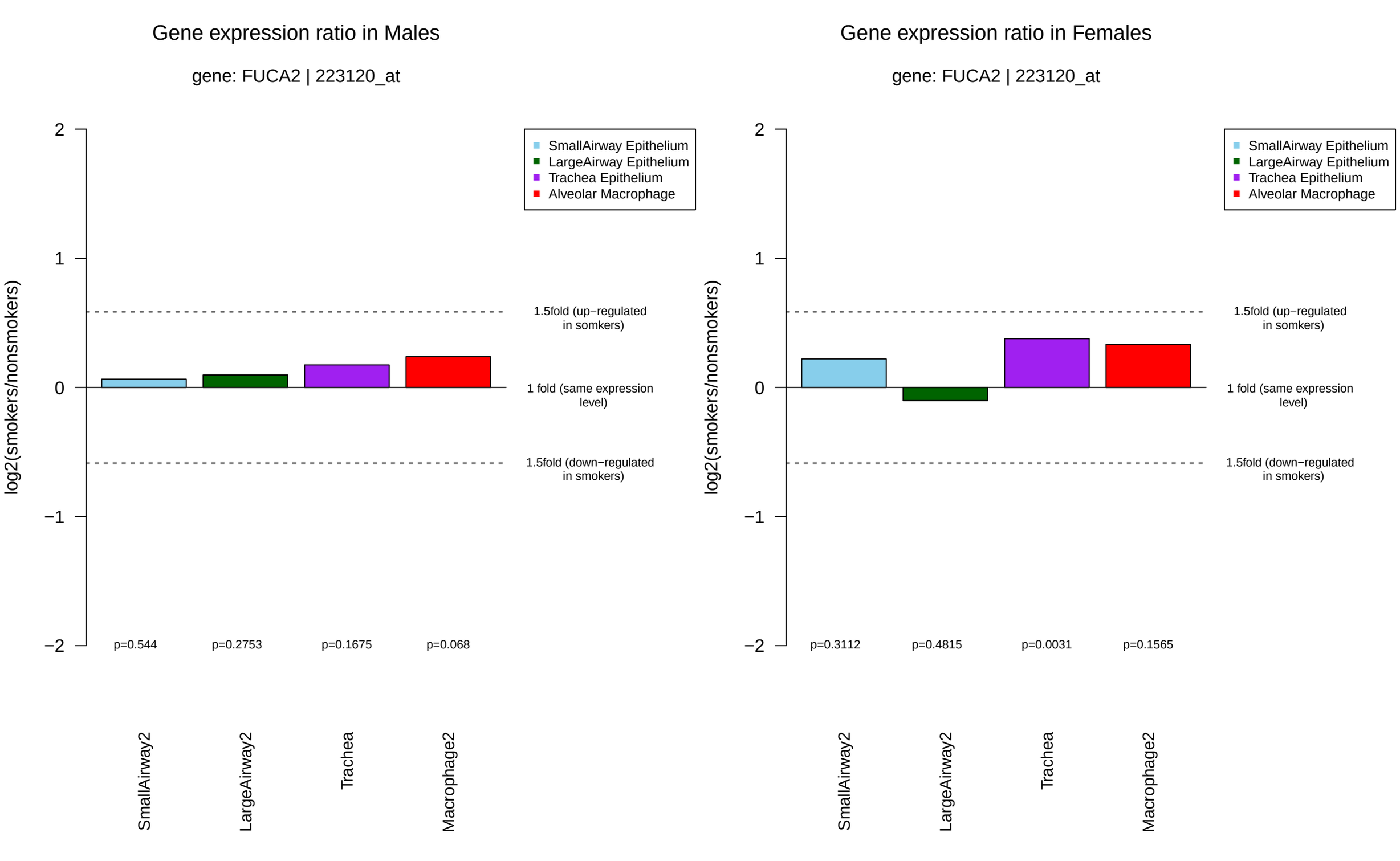
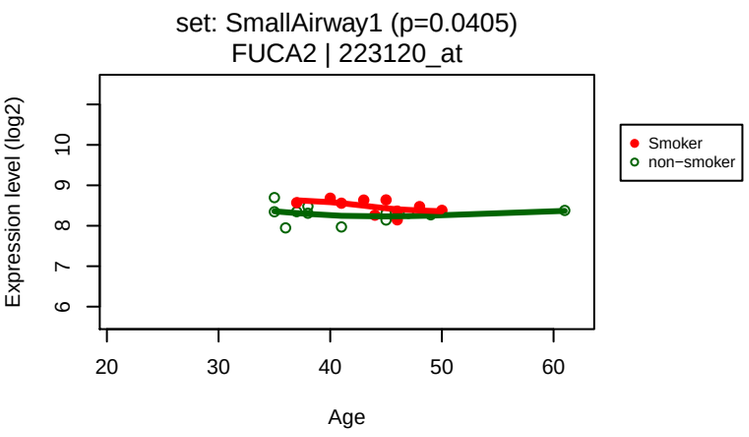
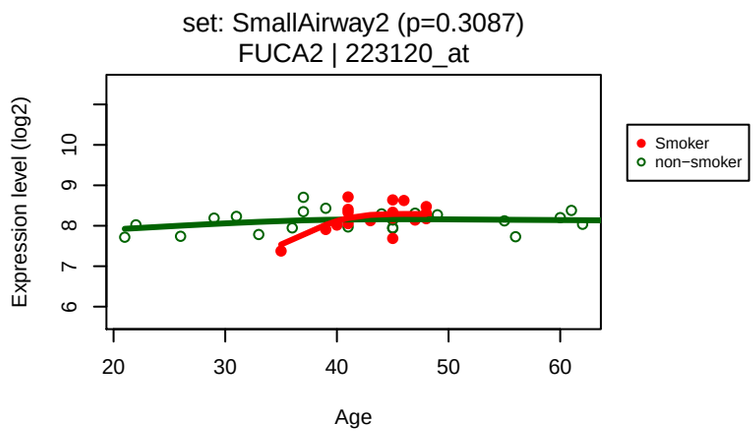
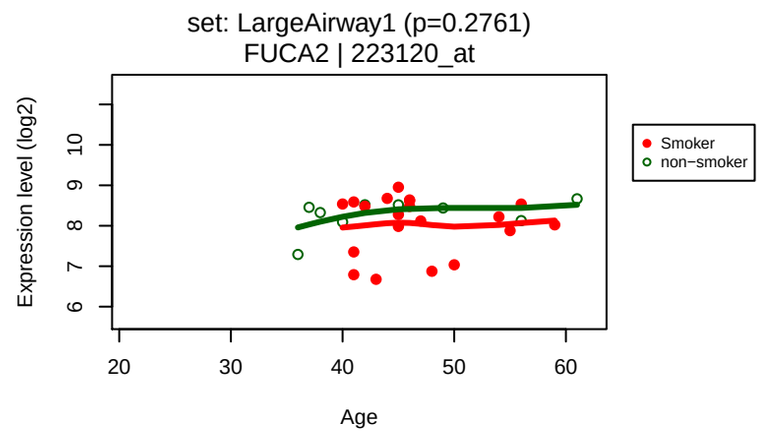

Gene expression of smokers and nonsmokers
gene: FUCA2 | 223120_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0405
p=0.3087
p=0.0399
p=0.2761
p=0.6798
p=0.0288
p=0.0838
p=0.0125
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

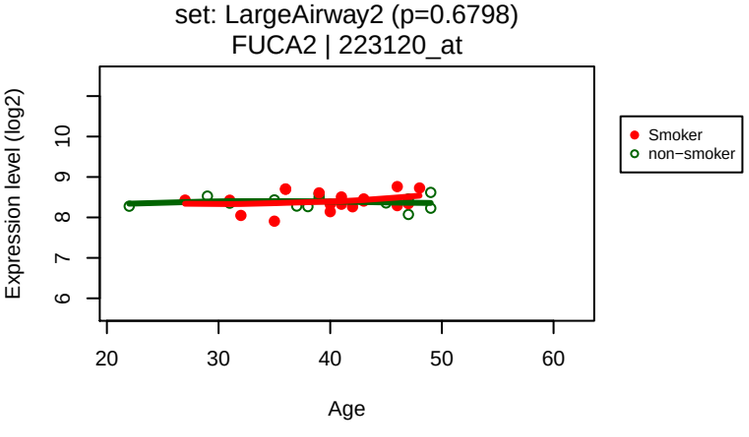

Gene expression ratio (smokers/nonsmokers)
gene: FUCA2 | 223120_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0405
p=0.3087
p=0.0399
p=0.2761
p=0.6798
p=0.0288
p=0.0838
p=0.0125
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

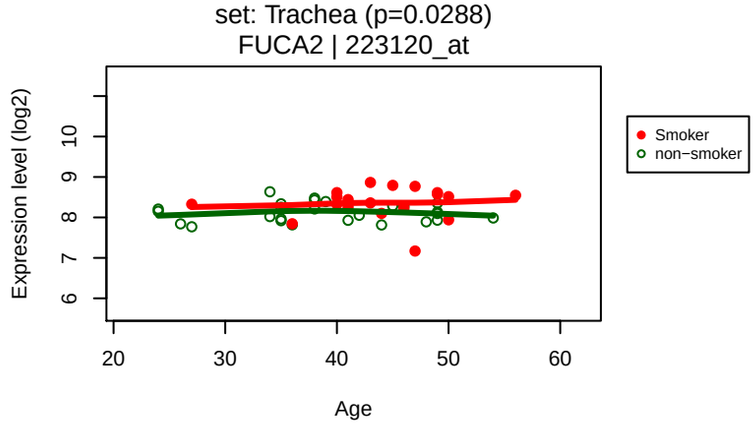

Gene expression ratio in Males
gene: FUCA2 | 223120_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.544
p=0.2753
p=0.1675
p=0.068
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: FUCA2 | 223120_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3112
p=0.4815
p=0.0031
p=0.1565
SmallAirway2
LargeAirway2
Trachea
Macrophage2

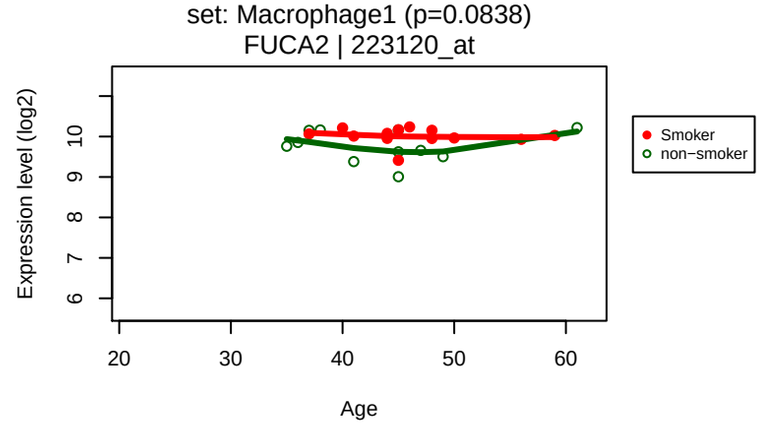

set: SmallAirway1 (p=0.0405)
FUCA2 | 223120_at
Expression level (log2)
Age
Smoker
non−smoker

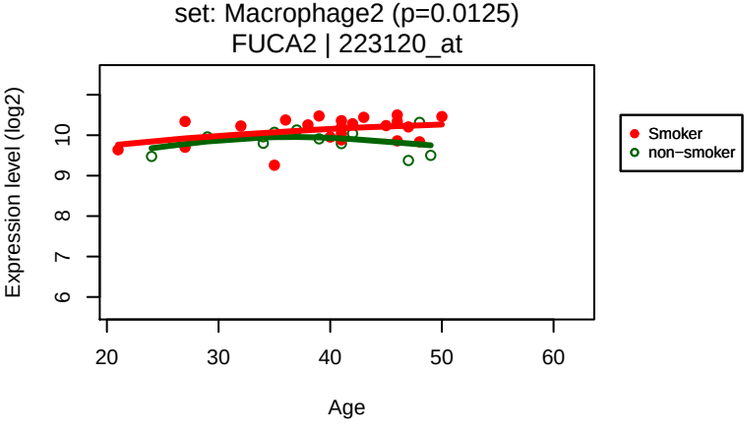

set: SmallAirway2 (p=0.3087)
FUCA2 | 223120_at
Expression level (log2)
Age
Smoker
non−smoker

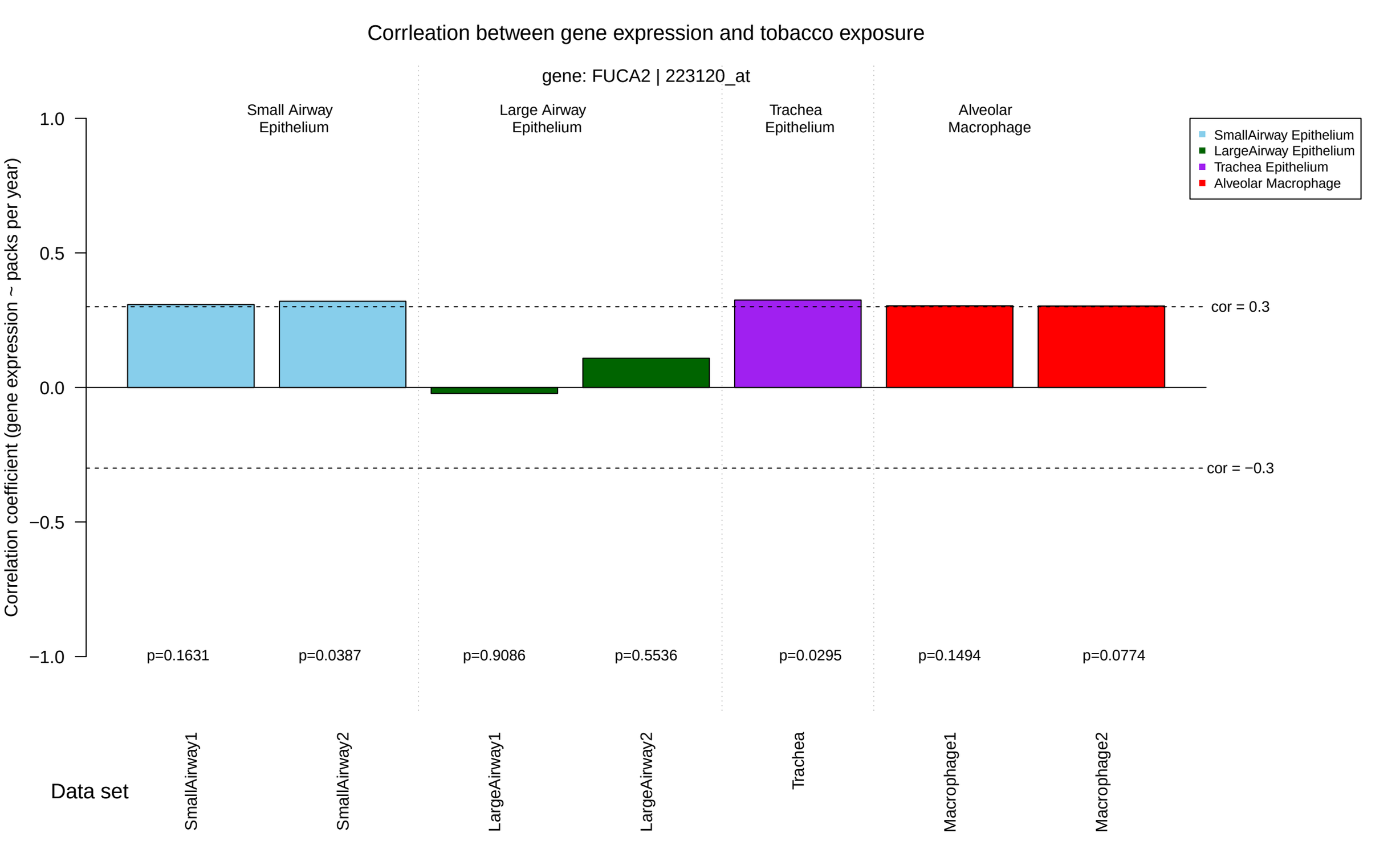

set: LargeAirway1 (p=0.2761)
FUCA2 | 223120_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6798)
FUCA2 | 223120_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0288)
FUCA2 | 223120_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0838)
FUCA2 | 223120_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0125)
FUCA2 | 223120_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: FUCA2 | 223120_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1631
p=0.0387
p=0.9086
p=0.5536
p=0.0295
p=0.1494
p=0.0774
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2